

Luke “Buck” Elliott

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EDUCATION

University of Texas at Austin – GPA: 3.85 / 4.00

May 2025

Bachelor of Science in Mechanical Engineering

Bachelor of Arts in Plan II Honors Program

- Selective multidisciplinary honors degree program admitting ~200 students yearly

Universitat Pompeu Fabra

May 2024 - June 2024

Mechatronics Engineering Study in Barcelona, Spain

- Study-abroad session which covered the theory & application of electronics within mechanical systems

WORK EXPERIENCE

Nanoassembly Lab – University of Texas – Undergraduate Researcher; Austin, TX

August 2023 – October 2024

- Assisted PhD candidates in subjecting DNA structures to a variety of thermal conditions and external stimuli
- Independently carried out novel test procedures and recorded experiment results for future analysis
- Analyzed resultant data and improved tests to achieve a higher yield of successful structures

Texas Instruments – Equipment Engineering Intern; Dallas, TX

May 2023 – August 2023

- Managed three long-term projects and improved reliability and output of existing semiconductor machinery
- Worked alongside professional engineers and technicians to implement continuous improvement initiatives
- Prepared weekly presentations and managed a GANTT chart to boost project progress and organization

PROJECTS

Failure Prediction w/ MEMS Ultrasonic Microphone | KiCad, Arduino

January 2025 – May 2025

- Utilized a low-cost MEMS microphone in the collection of ultrasonic frequencies indicating bearing failure
- Learned EE concepts such as PCB and circuit schematic design to enhance device efficiency and performance
- Analyzed resultant frequencies and crafted data plots using FFT to successfully determine faulty machinery

Machine Shop DIY Vice | CNC & Conventional Machining, Fusion 360, GD&T

January 2025 – May 2025

- Operated common machines – mill, lathe, CNC – to craft precisely toleranced steel & brass components
- Generated G-code, adhered to safety protocols, calculated proper MRR, and selected optimal machine tools
- Assembled toleranced components to create a working vice which met project specifications

Competition Battlebot | SolidWorks, Soldering, 3D-Printing

August 2024 – December 2024

- Led a small team of students to plan, design, and assemble a 3lb battlebot to compete with other student teams
- Used CAD software to design & model drivetrain, steering, and weaponry subsystems of the battlebot
- Learned & utilized key engineering concepts such as DFM, DFA, and FMEA to design a competitive product

VOLUNTEER AND LEADERSHIP EXPERIENCE

Phi Kappa Psi – Scholarship Committee; Austin, TX

December 2022 -Present

- Dedicated 4+ hours per week to mentoring and educating members presently struggling in their studies
- Planned, managed, and monitored weekly “study-hours” to foster an environment for academic engagement

EcoCAR EV Challenge – Connected Autonomous Vehicle Team; Austin, TX

December 2022 – April 2023

- Collaborated with a team of engineers to redesign and improve a complex vehicle assembly within Siemens NX
- Utilized Computer Aided Design (CAD) software to redesign and manage a collection of 500+ part files

SKILLS AND CERTIFICATIONS

Design Software:	SolidWorks, Autodesk Fusion360, KiCad
Technical Skills:	CAD, GD&T, Manual & CNC Machining, PCB Design, FMEA
Certifications:	Engineer in Training (EIT), CSWP (Certified SolidWorks Professional)
Coding Languages:	MATLAB, Python
Other Software	Microsoft Office (Word, PowerPoint, Excel, etc.)
Hobbies / Interests:	Lifting, Distance Running, Guitar, Go